100 eV is a high energy in comparison to ionization energies (usually <15 eV) and to the energies involved in chemical bonds (normally 1−5 eV). Therefore, nuclear radiation can cause several ionizations in its passage through matter; this is reflected in the common name *ionizing radiation*. Neutrons of energies <100 eV are included because their absorption (capture) by nuclei results in emission of nuclear radiation with energies much greater than 100 eV.

The passage of such high energy radiation through matter results in the transfer of energy to the atoms and molecules of the absorber material. This transfer of energy continues until the impinging particle of the radiation has reached the same average kinetic energy as the atoms comprising the material; i.e. until *thermal equilibrium* is obtained.

In considering the absorption of nuclear radiation it is appropriate to view the overall process from two aspects: (1) from the impinging particles point of view, i.e.processes occurring to the nuclear particles themselves as their energies are reduced to the thermal equilibrium value; such *absorption processes* are the principal consideration of this chapter; (2) from the absorbing materials point of view, i.e. processes in the absorbing material due to the effect of the transfer of energy. This transfer results initially in excitation and ionization which cause physical and chemical changes. The study of these effects is the domain of *radiation chemistry* and is considered in Chapter 8.

7.1. SURVEY OF ABSORPTION PROCESSES

The reduction in the intensity of a beam of ionizing particles can be caused either by reaction with the nuclei of the absorbing material (nuclear reactions) or with the atomic electrons (electron collision). In Table 7.1 the most important processes involved in the absorption of nuclear radiation in matter are listed along with the probability for each process. Comparison shows that the probability of interactions with electrons is considerably greater than that of a nuclear reaction; the only exception to this is the case of neutron absorption. In fact the principal mode of interaction between the particle and the atoms of the absorbing material involves the electromagnetic fields of the particle and the atomic electrons. Since neutrons are neutral particles which do not with electrons, in order for them to transfer energy to an atom it is necessary that they experience a direct collision with a nucleus. Consequently for all particles except neutrons, nuclear reactions can be neglected in considering the processes involved in the reduction of the intensity of the particle beam.

As nuclear radiation passes the atoms of an absorber, it can transfer some of its energy to the atomic electrons. If the amount of energy transferred is sufficient, ionization of the atom results. The positive ion and the electron thus formed are known as an *ion pair*. Frequently the electrons from this primary ionization have sufficiently high kinetic energy to cause *secondary ionization* in other atoms. The number of electrons produced in

Table 7.1 Survey of nuclear radiation absorption processes
The reaction cross-sections (σ) given are only the order of magnitude at about 1 MeV in Z ≈ 20

	Reacting particles and fields	Type of reaction	σ (barns)	Name of process
1	*Protons and heavier ions* react with			
1a	orbital electrons	Particle energy loss through atomic excitation and ionization	$\leq 10^5$	Ionization, (atomic) excitation
1b	atomic nucleus	Particle elastically scattered	≤ 10	Nuclear scattering
1c	atomic nucleus	Particle inelastically scattered	<1	Nuclear (coulomb) excitation
1d	atomic nucleus	Particle captured, formation of compound nucleus ($E_p > E_{cb}$(min)[c]	≤ 0.1	Nuclear transmutation
2	*Electrons* (e^-,β^-,β^+) react with			
2a	orbital electrons	Particle energy loss through atomic excitation and ionization	$>10^2$	Ionization, (atomic) excitation
2b		Slow β^+ annihilated, 2–3 photons formed	(100%)	Positron annihilation
2c	electric field of nucleus	Particle scattered with energy loss, continuous emission of hν (E_e »1 MeV)	>1	Bremsstrahlung
3	*Photons* (γ) react with			
3a	field of orbital electrons	γ scattered without energy loss	≤ 0.01	Coherent scattering
3b	free (outer) electrons	γ scattered with energy loss, ionization		Compton effect
3c	bound (inner) electrons	γ completely absorbed, one electron knocked out	≤ 10[a]	Photo effect
3d	field of nuclear force	γ annihilated, formation of positron-negatron pair ($E > 1.02$ MeV)		Pair formation
3e	atomic nucleus	γ scattered without energy loss	$\leq 10^{-3}$	Mössbauer effect
3f	atomic nucleus	γ scattered with energy loss		Nuclear excitation
3g	atomic nucleus	γ absorbed by nucleus, nuclear transmutation ($E > 5$ MeV)[b]		Nuclear photo effect
4	*Neutrons* react with			
4a	atomic nucleus	n scattered with energy loss	≤ 10	Neutron moderation
4b	atomic nucleus	n captured, nuclear transformation	$\leq 10^4$	Neutron capture

[a] See Fig. 7.17; σ increases strongly with decreasing energy.
[b] Threshold energy for Be(γ,α)^{4}He 1.6 MeV, D(γ,n)H 2.2 MeV.
[c] E_{cb}(min) is the Coulomb barrier energy, eqn. (10.15).

secondary ionization is often larger than that of the primary ionization but the average kinetic energies of the secondary electrons are lower than those of the primary electrons. In many interactions the initial radiation transfers insufficient energy for ionizations; instead an electron is raised to a higher, excited energy level of the atom. These *excited atoms* rapidly return to lower energy states by emission of electromagnetic radiation such as X-rays, visible light, etc. For neutrons the absorption process involving the capture of the neutron imparts sufficient recoil energy to cause ionization and excitation.

7.2. ABSORPTION CURVES

In order to measure the absorption of nuclear radiation, the experiments must be performed in such a manner as to eliminate as many of the interfering factors as possible. Usually a well-collimated beam is used. This is illustrated in Figure 7.1 for a point radioactive source. The relation between the disintegration rate A and the count rate R is given by Eqn (7.1):

$$R = \psi A \tag{7.1}$$

The *counting efficiency* ψ includes a number of factors:

$$\psi = \psi_{\text{sample}}\, \psi_{\text{abs}}\, \psi_{\text{det}}\, \psi_{\text{geom}} \tag{7.2}$$

If conditions were ideal, there would be no self-absorption or scattering in the sample (in which case $\psi_{\text{sample}} = 1$), no absorption of radiation between the sample and the detector window ($\psi_{\text{abs}} = 1$), and the detector would have a 100% efficiency (sensitivity) to a "count" for each particle reaching its window ($\psi_{\text{det}} = 1$).

The geometric efficiency ψ_{geom}, is 1 for 4π-geometry, i.e. for a spherical detector subtending a 360° solid angle about the sample. Although such detectors exist (Ch. 9), more commonly the sample is counted outside the detector at some distance r, as indicated in Figure 7.1. If the detector window offers an area of S_{det} perpendicular to the radiation, the geometrical efficiency is approximated by (for small ψ_{geom})

$$\psi_{\text{geom}} \approx S_{\text{det}} / \left(4\pi r^2\right) \tag{7.3}$$

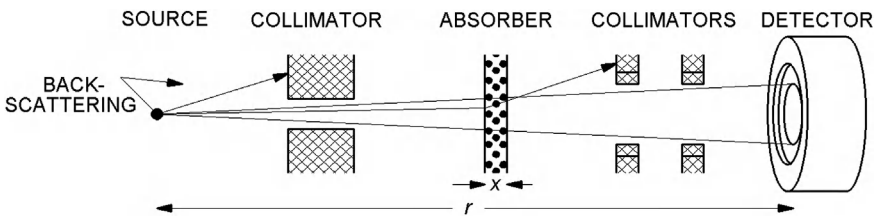


Figure 7.1 Geometrical arrangement for measuring absorption curves.